NATIONAL UNIVERSITY OF COMPUTER AND EMERGING SCIENCES
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ABSTRACT

The Quiz Game is a console-based application developed using the C programming language. It is designed to allow users to test their knowledge by answering multiple-choice questions. The program displays questions along with different answer options and checks the user's selected answers. It keeps track of the score and displays the final result after all questions have been answered. The application will also include an option to replay the quiz. This report explains the background, specifications, design, implementation, and testing of the Quiz Game project.

INTRODUCTION

The Quiz Game is a simple interactive application designed to test a user's knowledge through multiple-choice questions. It provides questions with several answer options, from which the user selects one answer. The program checks the selected option and determines whether it is correct or incorrect. It also maintains the user's score throughout the quiz and displays the final result at the end. The project will be developed using the C programming language and basic programming constructs such as variables, loops, conditional statements, arrays, and functions. The main goal is to apply fundamental programming concepts in a practical application.

BACKGROUND (RESEARCH AND PROJECT SELECTION)

The idea for this project came from the common use of quiz applications for learning and testing knowledge. Before selecting the project, we considered simple console-based applications that could be developed using fundamental C programming concepts. We selected the Quiz Game because it provides an opportunity to create an interactive program while practising basic programming logic.

This project brings together several programming techniques:

- **Variables:** Used to store user answers, question data, and the score.
- **Conditional Statements:** Used to check answers and determine whether they are correct.
- **Loops:** Used to display questions repeatedly and control the quiz flow.
- **Arrays:** Used to organize and store questions and their answer options.
- **Functions:** Used to divide the program into smaller and manageable parts.
- **User Input:** Used to receive answers from the user.

The Quiz Game was chosen because it is practical, understandable, and suitable for a beginner-level C programming project. It reinforces important course concepts and helps improve logical thinking and problem-solving skills.

PROJECT SPECIFICATIONS

FEATURE	DESCRIPTION
Start Quiz	Starts the quiz and displays the questions.
Multiple-Choice Questions	Displays each question with multiple answer options.
Answer Validation	Checks whether the user's selected answer is correct or incorrect.
Score Calculation	Keeps track of the number of correct answers.
Final Result	Displays the user's final score after completing the quiz.
Replay Option	Allows the user to play the quiz again.

The following functions are proposed for the development of the Quiz Game.

PROCEDURE / FUNCTION NAME	PURPOSE
`displayQuestions()`	Displays the quiz questions and their options.
`checkAnswer()`	Checks whether the user's answer is correct.
`calculateScore()`	Updates and maintains the user's score.
`displayResult()`	Displays the final score and result.
`startQuiz()`	Controls the main quiz process.

PROBLEM ANALYSIS

PROBLEM	IMPACT	ANALYSIS	SOLUTION
Invalid user input	May cause incorrect answer processing.	The user may enter an option outside the available choices.	Validate the entered option before checking the answer.
Incorrect answer checking	May produce an incorrect score.	The selected answer must be compared with the correct answer.	Use conditional statements to compare answers.
Score management	May result in an incorrect final score.	The score must be updated only when an answer is correct.	Initialize the score and update it using conditional statements.
Repeated questions	The quiz may not progress correctly.	Questions must be displayed in the correct sequence.	Use loops to control the question sequence.
Replay functionality	The next game may show an incorrect score.	The previous score must not carry over into a new game.	Reset the score when a new game starts.

SOLUTION DESIGN

1. [Design Aspect Title]

1. User Interface Design

The project will use a simple console-based interface. It will display instructions, questions, answer options, and the final score in a clear format.

- Display the quiz questions one by one.
- Show multiple answer options for each question.
- Allow the user to enter an answer through the keyboard.

2. Question Management

The questions and their answer options will be organized using arrays. This will help keep the question data structured and make it easier to display questions in sequence.

- Store questions and answer options.
- Maintain the correct answer for each question.
- Display questions in an organized order.

3. Answer Validation

The program will compare the user's selected answer with the correct answer. Conditional statements will be used to determine whether the answer is correct or incorrect.

- Receive the user's selected option.
- Compare it with the correct answer.
- Display appropriate feedback.

4. Score Management

A score variable will be used to maintain the number of correct answers. The score will increase whenever the user provides a correct answer.

- Initialize the score at the beginning of the quiz.
- Update the score after each correct answer.
- Display the final score at the end.

5. Replay Functionality

The program will provide an option to play the quiz again after displaying the final result.

- Ask the user whether they want to replay.
- Reset the score for a new game.
- Start the quiz again if the user chooses to replay.

IMPLEMENTATION AND TESTING

The Quiz Game will be developed using the C programming language. The program will use the standard `stdio.h` library for input and output operations. Its structure will be divided into functions to make the code easier to understand and manage. Arrays will be used to organize questions and answer options, while loops and conditional statements will control the quiz process. User input will be checked to determine whether the selected answers are correct. Testing will be performed after development to check the program's main features and ensure that the score and final result are calculated correctly.

The planned testing activities include:

- **Input Testing:** Checking whether the program accepts the user's selected answer.
- **Answer Validation Testing:** Checking whether correct and incorrect answers are identified properly.
- **Score Testing:** Confirming that the score increases only for correct answers.
- **Result Testing:** Checking whether the final score is displayed correctly.
- **Replay Testing:** Checking whether a new game starts with a reset score.

PROJECT BREAKDOWN STRUCTURE

The project will be completed in the following phases:

- **Phase 1 – Planning:** Selecting the project idea and preparing the project proposal.
- **Phase 2 – Design:** Planning the program structure, features, questions, and functions.
- **Phase 3 – Development:** Writing the C program and implementing the planned features.
- **Phase 4 – Testing:** Testing the program, identifying errors, and correcting them.
- **Phase 5 – Documentation:** Preparing the final project documentation and organizing screenshots of the program.

Progress will be monitored by completing each phase in sequence. Regular reviews will be conducted to ensure that the project is progressing according to the plan.

IMPLEMENTATION SCREENSHOTS AND OUTPUT

Screenshots of the completed program, including the quiz questions, answer validation, final score, and replay option, will be included after the development of the project.

CONCLUSION

The Quiz Game project aims to provide a simple and interactive way to test knowledge through multiple-choice questions. It will apply fundamental C programming concepts such as variables, arrays, loops, conditional statements, and functions. Through this project, we expect to improve our programming skills, logical thinking, and problem-solving abilities. It will also provide practical experience in developing a console-based application.